$$\frac{d^2y}{dt^2} = f\left(\frac{dy}{dt}, y, t\right).$$

1° homogenous

□ existence & uniqueness thm.

if $y'' + p(t)y' + q(t)y = 0$, $y(t_0) = y_0$, $y'(t_0) = y_0'$.

where p, q are continuous on an open interval I that contain t_0. There exists one exact solution $y = \phi(t)$ throughout the interval.

□ superposition.

if y_1, y_2 are the two solutions of $L(y)$, then $c_1y_1 + c_2y_2$ is as well.

proof: $L[c_1y_1 + c_2y_2] = (c_1y_1 + c_2y_2)'' + p(t)\cdot(c_1y_1 + c_2y_2)' + q(t)\cdot(c_1y_1 + c_2y_2)$

$$= c_1\cdot(y_1'' + p(t)\cdot y_1' + q(t)\cdot y_1) + c_2\cdot(y_2'' + p(t)\cdot y_2' + q(t)y_2)$$

$$= c_1L[y_1] + c_2L[y_2] = 0.$$

Since $L[y_1]$, $L[y_2] = 0$

Also, we know that $y(t_0) = y_0$, $y'(t_0) = y_0'$

$$\begin{cases} c_1y_1(t_0) + c_2y_2(t_0) = y_0 \\ c_1y_1'(t_0) + c_2y_2'(t_0) = y_0' \end{cases} \Rightarrow \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} y_0 \\ y_0' \end{bmatrix}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_1'y_2 \neq 0 \Rightarrow c_1y_1 + c_2y_2 \text{ is a solution}$$

By Cramer's rule,

$$c_1 = \frac{\begin{vmatrix} y_0 & y_2 \\ y_0' & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}, \quad c_2 = \frac{\begin{vmatrix} y_1 & y_0 \\ y_1' & y_0' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

1] complex-valued solutions

if $y = u(t) + iv(t)$ is a solution, then real part $u(t)$ and its imaginary part $v(t)$ are also solutions of the equation.

$$L[y](t) = (u(t) + iv(t))'' + p(t)\cdot(u(t) + iv(t))' + u(t) + i\cdot v(t)$$

$$= [u''(t) + p(t)\cdot u'(t) + u(t)] + i(v''(t) + p(t)\cdot v'(t) + v(t))$$

$$= L[u](t) + i\,L[v](t) = 0$$

$$\Rightarrow L[u] = 0, \quad L[v] = 0$$

1] Abel's Theorem

$$W[y_1, y_2](t) = c\exp\left(-\int p(t)\,dt\right)$$

proof: $y_1'' + p(t)y_1' + y_1 = 0$ ①

$y_2'' + p(t)y_2' + y_2 = 0$ ②

① $\times y_2$ − ② $\times y_1$ ⇒

$$(y_1''y_2 + p(t)y_1'y_2 + y_1y_2) - (y_2''y_1 + p(t)y_2'y_1 + y_1y_2) = 0$$

$$\Rightarrow (y_1''y_2 - y_2''y_1) + (y_1'y_2 - y_2'y_1)p(t) = 0$$

$$W' = y_1''y_2 + y_1'y_2' - y_2''y_1 - y_2'y_1' = y_1''y_2 - y_2''y_1$$

$$\Rightarrow W' + W p(t) = 0.$$

$$\Rightarrow W = c\cdot e^{-\int p(t)\,dt}$$

Q3. $W = \begin{vmatrix} e^{-2t} & te^{-2t} \\ -2e^{-2t} & (1-2t)e^{-2t} \end{vmatrix} = (1-2t)\cdot e^{-4t} + 2te^{-4t} = e^{-4t}$

Q4. $W = \begin{bmatrix} e^{t}\sin t & e^{t}\cos t \\ e^{t}(\sin t+\cos t) & e^{t}(\cos t-\sin t) \end{bmatrix} = e^{2t}\cdot(sc - s^2) - e^{2t}\cdot(sc + c^2) = e^{2t}\cdot(-s^2-c^2)$

$= -e^{2t}$

Q5. $W = \begin{bmatrix} \cos^2\theta & 1+\cos 2\theta \\ -2\cos\theta\sin\theta & -2\sin 2\theta \end{bmatrix} = \cos^2\theta\cdot(-2\sin 2\theta) + (1+\cos 2\theta)\cdot\sin 2\theta$

$= \sin 2\theta\cdot(\cancel{-2\cos^2\theta + 1} + \cancel{2\cos^2\theta - 1})$

$= 0$

Q10. $t^2y'' - 2y = 0 \Rightarrow y'' - \frac{2}{t^2}y = 0$, $t > 0$

$r^2 - \frac{2}{t^2} = 0 \Rightarrow r = \pm\frac{2}{|t|}$

$W = \begin{vmatrix} t^2 & \frac{1}{t} \\ 2t & -\frac{1}{t^2} \end{vmatrix} = -1 - 2 = -3 \neq 0$ ✓.

Q16: $W[y_1, y_2] \neq 0$.

if $a_1b_2 - a_2b_1 \neq 0$, then they are.

Q23. $t^2y'' - t(t+2)y' + (t+2)y = 0$

$\Rightarrow y' - \frac{t(t+2)}{t^2}y' + \frac{t+2}{t^2}y = 0$

$W[y_1, y_2] = c\cdot e^{-\int -\frac{t(t+2)}{t^2}dt}$

$= c\cdot e^{\int 1+\frac{2}{t}dt}$

$= c\cdot e_2^{t+2\ln|t|}$

$= ct^2\cdot e^{t}$. $t \neq 0$.

complex roots

proof: $y = c_1 \cdot e^{(\lambda + i\mu)t} + c_2 \cdot e^{(\lambda - i\mu)t}$

$r_1 = \lambda \pm i\mu$.

Euler's ∽: $e^{it} = \cos t + i \sin t$

$\Rightarrow e^{\lambda t} \cdot (k_1 \cos(\mu t) + k_2 \cos(\mu t))$

实虚部可做solution了

$y = e^{\lambda t} \cdot (c_1 \cos(\mu t) + c_2 \sin(\mu t)) = e^{\lambda t}(c_1 \cos(\mu t) + c_2 \sin(\mu t))$

if $r_1 = r_2 = c$

$y = (c_1 + c_2 t) e^{rt}$

□ reduction of Order

variation of parameter

$$y'' + p(t)y' + q(t)y = g(t)$$

$$\Rightarrow \quad y'' + p(t)y' + q(t)y = 0$$

$$\Rightarrow \quad r^2 + p \cdot r + q = 0$$

$$\Rightarrow \quad y_1 \ , y_2$$

$$y = u_1(t)\, y_1(t) + u_2(t) \cdot y_2(t)$$

$$y' = y_1'(t)\, u_1(t) + y_2'(t)\, u_2(t) + \underbrace{y_1(t) \cdot u_1'(t) + y_2(t)\, u_2'(t)}_{0}$$

$$y'' = u_1'(t)\, y_1'(t) + y_1''(t)\, u_1(t) + y_2''(t)\, u_2(t) + y_2'(t)\, u_2'(t)$$

$$y'' + py' + q = u_1(t)\,(\underbrace{y_1''(t) + p(t)\, y_1'(t) + q(t)\, y_1(t)}_{0})$$

$$+ \ u_2(t)\,(\underbrace{y_2''(t) + p(t) \cdot y_2'(t) + q(t)\, y_2(t)}_{0}) + u_1'(t)\, y_1'(t)$$

$$+ \ y_2'(t)\, u_2'(t) = g(t)$$

homogeneous

$$\begin{cases} u_1'(t)\, y_1(t) + u_2'(t)\, y_2(t) = 0 \\ u_1'(t)\, y_1'(t) + y_2'(t)\, u_2'(t) = g(t) \end{cases}$$

$$\Rightarrow \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ g \end{pmatrix}$$

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ g & y_2' \end{vmatrix}}{W[y_1, y_2]} = \frac{-g(t) \cdot y_2}{W[y_1, y_2]} \qquad u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & g \end{vmatrix}}{W[y_1, y_2]} = \frac{g(t) \cdot y_1}{W[y_1, y_2]}$$

$$u_1(t) = \int \frac{-g(t) \cdot y_2(t)}{W[y_1, y_2](t)}\, dt \ , \quad u_2(t) = \int \frac{g(t) \cdot y_1(t)}{W[y_1, y_2](t)}\, dt$$

$$Y(t) = -y_1(t)\int \frac{g(t)\cdot y_2(t)}{W[y_1, y_2](t)}\,dt + y_2(t)\int \frac{g(t)\cdot y_1(t)}{W[y_1, y_2](t)}\,dt$$

$$y = c_1 y_1(t) + c_2 y_2(t) + Y(t)$$

e.g.1 $y'' + y = \tan t \quad . \quad 0 < t < \pi/2$

$r^2 + 1 = 0$

$r = \pm i$

$y_1(t) = \sin t \quad . \quad y_2(t) = \cos t$

$\frac{1-c^2}{c}$.

sec

$$W[y_1, y_2] = \begin{vmatrix} \sin t & \cos t \\ \cos t & -\sin t \end{vmatrix} = -1$$

$$Y(t) = -\sin t \cdot \int \frac{\cos t \cdot \tan t}{-1}\,dt + \cos t \cdot \int \frac{\tan t \cdot \sin t}{-1}\,dt$$

$$= \sin t \cdot (-\cos t) - \cos t \cdot (\ln(\sec t + \tan t) - \sin t)$$

$$= \cancel{-\sec} - \cos t \cdot \ln \cos t + \cancel{\cos}$$

$$= -\cos t \cdot \ln|\cos t|$$